

Does shore power cut harbour pollution by 90%?

Testing a public claim about the Grand Harbour shore-to-ship project against connection records, EU law and research

THE CLAIM UNDER REVIEW

“This initiative promises to slash 90% of air pollution in the Grand Harbour.”

Infrastructure Malta, Shore-to-Ship project page, 1 December 2023.
Repeated at the inauguration, 10 July 2024.

VERDICT: MISLEADING

A design potential presented as a result; ships plugged in 9% of the time

Version 1.2 · 5 October 2026

Status: draft (right of reply: Infrastructure Malta, Transport Malta)

Prepared from public sources and published records.

Repository: github.com/leandergrech/Mizien

MIZIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

Infrastructure Malta’s project page for the Grand Harbour shore-to-ship system says the initiative “**promises to slash 90% of air pollution in the Grand Harbour**”. At the July 2024 inauguration the government said its completion meant a 90% cut and cleaner air for 17,000 families. We tested the figure against how the system is actually used, the law that governs it, and peer-reviewed research on cruise emissions in port.

- 1

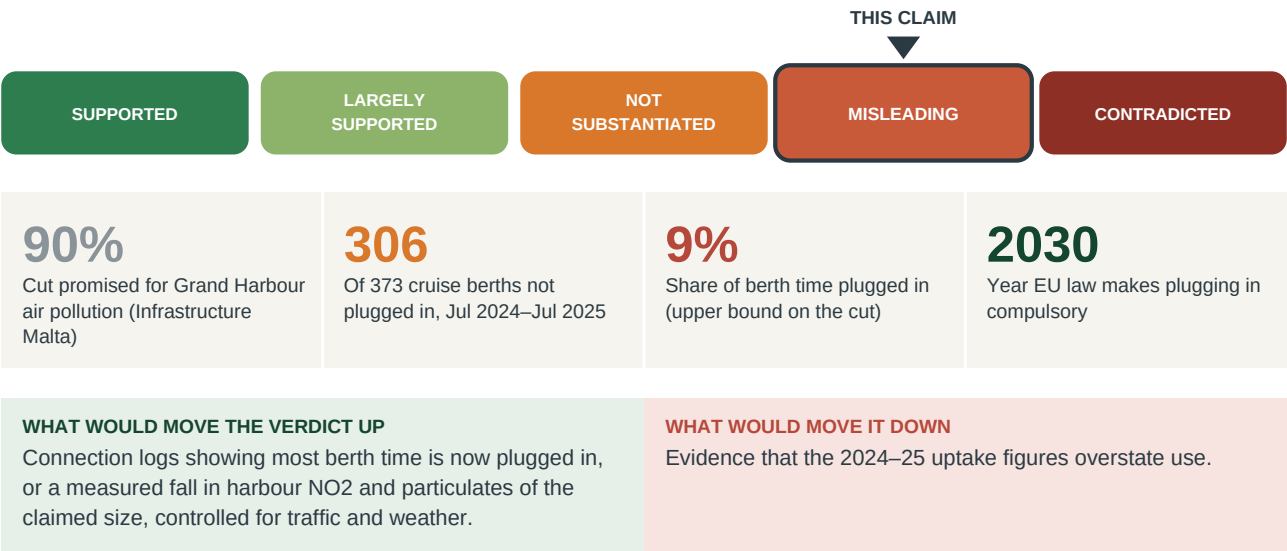
Shore power works for a ship that plugs in. Infrastructure Malta’s own figures, published since 2020, are for a liner that switches off its auxiliary engines: 93% less nitrogen dioxide, 92.6% less particulate matter, 99.6% less sulphur dioxide (39.6% less CO2). Its earlier pages gave the 90% as a cut in the pollution ships produce; the 2023 page dropped that scope.
- 2

But plugging in is voluntary until 2030. EU law (FuelEU Maritime, Art. 6) only makes connection compulsory from 1 January 2030. Transport Malta says connecting “remains the prerogative of the shipping line”. Neither the project page nor the launch said so.
- 3

In its first year, ships plugged in for 9% of their time at berth. Records obtained by Amphora Media from Transport Malta show 67 connections in 373 berths, July 2024 to July 2025; 306 were not plugged in. One LNG-powered ship made 35 of the 67; other ships plugged in at one berth in ten.
- 4

Cruise traffic grew. Calls rose 8%, from 357 in 2024 to 385 in 2025, about as much as the 9% of berth time plugged in from July 2024 to July 2025, so extra traffic may offset much of the first-year saving. The periods differ, and shore power was available from July 2024. No air-quality study has measured the effect in the harbour.
- 5

Verdict: misleading (moderate confidence). The 90% holds per connected ship on Infrastructure Malta’s own figures, but the 2023 page presented it as a cut in Grand Harbour air pollution and the launch as a result of completing the project. That omits the voluntary connection and actual use, which the evidence puts nearer a tenth of the claim.



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1 The claim and what we could verify	5 Where the evidence points different ways
2 Method	6 Testing the claim
3 What the research says	7 Verdict and requests for evidence
4 How the system is used	8 Limitations

1

The claim and what we could verify

The claim is on Infrastructure Malta’s project page, published on 1 December 2023 when the first ships were connected in a trial [1]. It was repeated at the inauguration on 10 July 2024, reported by TVM, the public broadcaster [2], and by MaltaToday [3]. The project page was read in full; the press conference is known through reported speech.

What was said	Who	Access
“This initiative promises to slash 90% of air pollution in the Grand Harbour.”	Infrastructure Malta [1]	Read in full
Completion “means a drastic reduction of 90% of the pollution generated by ships entering the Grand Harbour”	Press conference, as reported [2]	Reported speech
17,000 families around the harbour “will directly benefit” from cleaner air	Prime Minister, as reported [2]	Reported speech
90% when five liners use the system at once	Press conference, as reported [3]	Summary read

Earlier wording. Infrastructure Malta’s pages on the project from 2020 to 2023 gave the figure a narrower scope. On 30 November 2020 it said the project “will cut over 90% of the air pollution that cruise liners and Ro-Ro ships produce when visiting Malta’s principal port”, and that by switching off their auxiliary engines cruise liners would emit 93% less nitrogen dioxide, 92.6% less particulate matter, 99.6% less sulphur dioxide and 39.6% less carbon dioxide [10]. A page of 19 February 2022 repeats these figures [11]. The 2023 project page dropped the scope: “90% of air pollution in the Grand Harbour”.

SCOPE NOTE

This check is about the 90% figure. It does not question that shore power is worth building, and it does not assess the electricity supply, the grid outages alleged in 2025 (denied by Transport Malta and Enemalta), or noise.

2

Method

Question. Has, or will, the shore-to-ship project cut Grand Harbour air pollution by 90%?

Evidence. The project page, earlier Infrastructure Malta pages and launch reports [1–3, 10, 11]; connection records for July 2024 to July 2025 obtained by Amphora Media from Transport Malta under freedom of information [4] (second-hand: the reply is not public); Valletta Cruise Port’s traffic figures [5]; Regulation (EU) 2023/1805 [6]; peer-reviewed studies found through Crossref and OpenAlex [7–9]; MSC Cruises’ description of its ship [12]. Numbers are recomputed by *tools/cc-016-report/calc.py* from *data/cc-016/*.

Grades. Peer-reviewed observational studies are grade B; official records and reviews grade C. ♦ marks data known second-hand. **Verdicts** follow the five-point scale in Appendix A.

3 What the research says

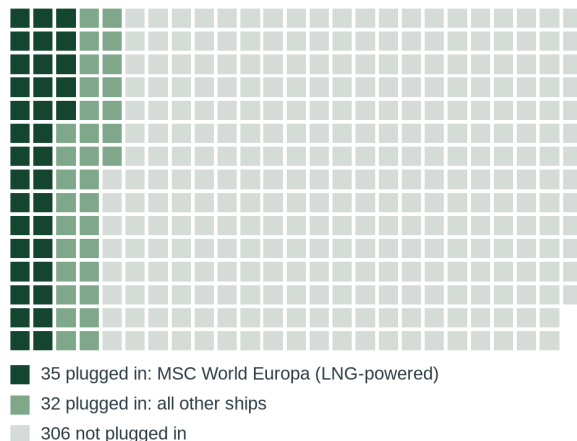
Infrastructure Malta's own figures are the basis of the 90%: a cruise liner that switches off its auxiliary engines at the quay emits 93% less nitrogen dioxide, 92.6% less particulate matter and 99.6% less sulphur dioxide, and 39.6% less carbon dioxide [10, 11]. The study behind these percentages is not cited. They apply to *a ship that connects*. Research is consistent with a large cut per ship: a bottom-up inventory of 164 cruise calls at Istanbul's Galataport in 2024 found that hotelling, the time at berth with auxiliary engines running, produced more than 90% of the ships' CO₂ in port and most of their NO_x and SO_x [7]. Emissions do not disappear: they move to the power station supplying the grid.

Shore power does nothing for manoeuvring. A sensor network in the Bay of Kotor found particulate spikes when cruise ships arrived and departed [8], and a 2025 review lists cost, ship retrofits and voluntary use among the barriers to adoption [9]. Pollution in the harbour also comes from ferries, tugs, other ships, road traffic and the power station; the claim speaks of all air pollution in the Grand Harbour.

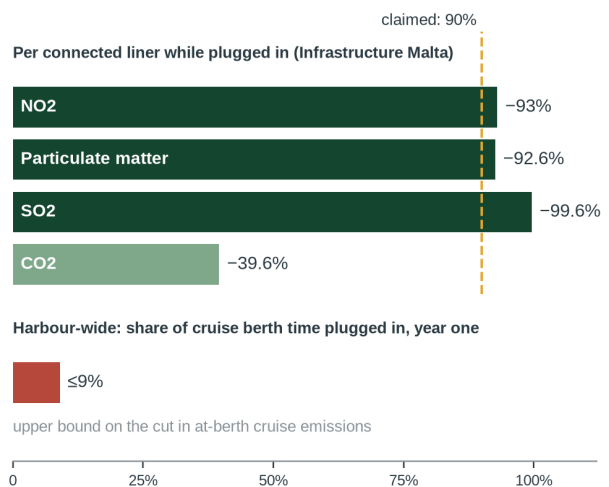
The ship that plugged in most often, MSC World Europa (Section 4), is powered by liquefied natural gas. Its owner describes it as "the world's largest LNG-powered cruise ship", with five dual-fuel engines that run on LNG or on low-sulphur marine gasoil, and says LNG "nearly eliminates" sulphur oxides and fine particles compared with standard marine fuels [12]. If it runs on LNG at the quay, plugging it in saves less sulphur dioxide and particulate matter than Infrastructure Malta's per-liner figures suggest. Which fuel it burned at berth in Valletta is not known.

4 How the system is used

The 373 cruise berths, Jul 2024 – Jul 2025 (one square each)

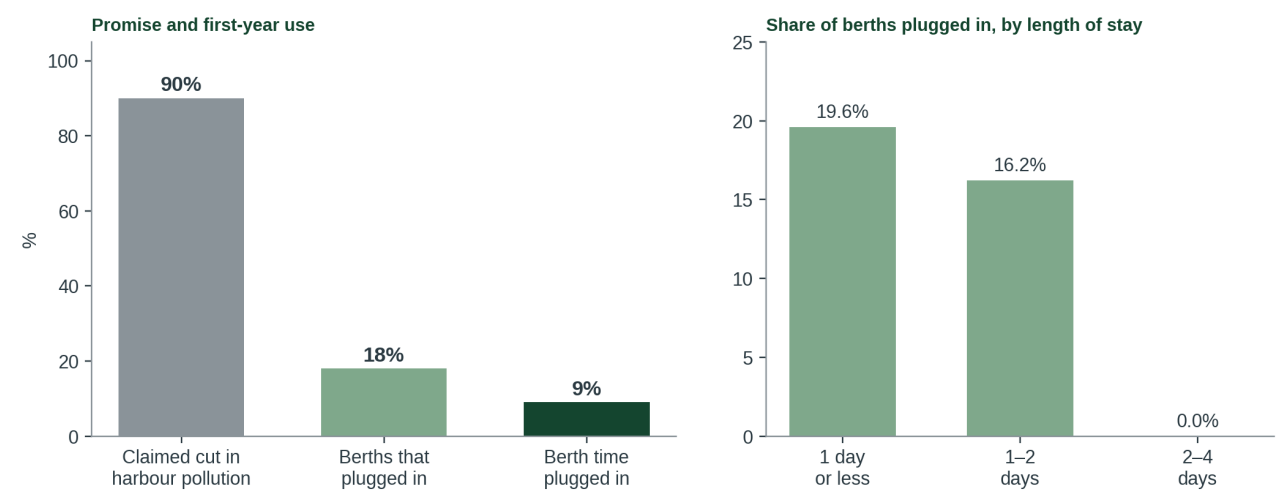


The same 90%, two scopes



Berth and connection counts: Amphora Media analysis of Transport Malta FOI records and the Valletta Cruise Port schedule, 10 Jul 2024 – 10 Jul 2025 (second-hand; the FOI reply is not public).
 Per-liner cuts: Infrastructure Malta, 30 Nov 2020 (study not cited). MSC World Europa's fuel: MSC Cruises, 24 Oct 2022 (dual-fuel LNG / low-sulphur marine gasoil).

Figure 1. Left: the 373 cruise berths from July 2024 to July 2025; 67 plugged in, 35 of them by MSC World Europa, and 306 did not. Right: the 90% holds for a connected liner on Infrastructure Malta's figures [10], but in the first year ships were plugged in for 9% of berth time, which caps the harbour-wide cut in at-berth cruise emissions at about 9%. Connection data second-hand ♦ [4].



10 Jul 2024 – 10 Jul 2025. Sources: Infrastructure Malta; Amphora Media analysis of Transport Malta FOI records and the Valletta Cruise Port schedule (second-hand).

Figure 2. Left: the promised cut against the share of berths and of berth time plugged in during the first year. Right: plug-in rate by length of stay. Data ♦ [4].

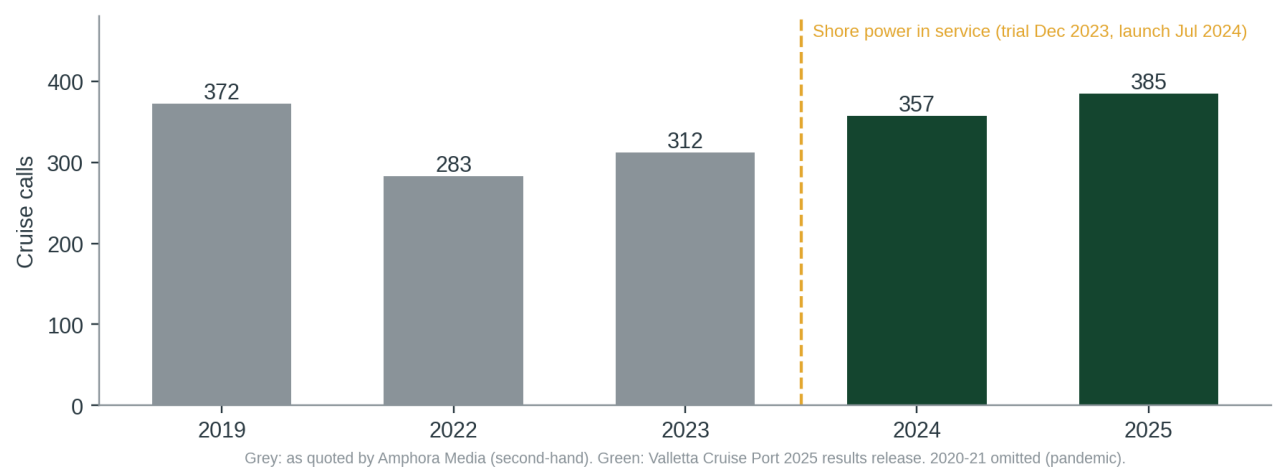


Figure 3. Cruise calls at Valletta Cruise Port [5]; 2019–2023 as quoted in [4]. Infrastructure Malta, citing the NSO, also gives 372 cruise liners for 2019 [10].

Indicator	Value	Source	Grade
Berths plugged in, Jul 2024–Jul 2025	67 of 373 (18%)	Transport Malta FOI ♦ [4]	C
Share of berth time plugged in	9%	Amphora Media analysis ♦ [4]	C
Connections by MSC World Europa	35 of 67 (52%)	Transport Malta FOI ♦ [4]	C
Plug-in rate: MSC World Europa; all other ships	35 of 51 (69%); 32 of 322 (10%)	Calculated from [4] ♦	C
Carnival Corporation calls plugged in	6 of 58	Transport Malta FOI ♦ [4]	C
Stays of 2–4 days plugged in	0%	Amphora Media analysis ♦ [4]	C
Cruise calls 2024 → 2025	357 → 385 (+8%)	Valletta Cruise Port [5]	C
Connection compulsory from	1 Jan 2030	Reg. (EU) 2023/1805 Art. 6 [6]	C

All values in data/cc-016/checks.csv. Transport Malta said Carnival’s connections were for testing and certification, after which its ships “will be able to use the system routinely” [4].

5

Where the evidence points different ways

Q1 Is 90% the right number for a connected ship?

LARGELY, FOR NO_x, PM, SO_x

EVIDENCE FOR THE CLAIM'S IMPRESSION

Infrastructure Malta's figures for a liner on shore power: 93% less NO₂, 92.6% less particulate matter, 99.6% less SO₂ [10, 11]. Hotelling dominates cruise emissions in port [7], so a large cut per ship is consistent with the research.

EVIDENCE AGAINST

These are Infrastructure Malta's estimates; the study behind them is not cited. They apply to ships that connect, not to all harbour pollution; CO₂ falls 39.6%; manoeuvring emissions remain [8]. For an LNG-fuelled ship the sulphur and particulate saving is smaller [12].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: the figure holds per connected ship for nitrogen dioxide, particulates and sulphur dioxide, which is how Infrastructure Malta's per-liner figures are stated.

Q2 Did completion of the project deliver 90%?

NO

EVIDENCE FOR THE CLAIM'S IMPRESSION

Use is rising: one large ship (MSC World Europa) plugged in at 35 of its 51 calls, and Carnival ships were certifying in 2025 [4].

EVIDENCE AGAINST

Ships were plugged in for 9% of berth time in the first year; with 100% removal while plugged in, the cut in at-berth cruise emissions could be at most about 9%. Calls rose 8% in 2025.

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: presented as a result of completion, the 90% overstates the first-year effect about tenfold.

Q3 Will it reach 90%?

NOT BEFORE 2030

EVIDENCE FOR THE CLAIM'S IMPRESSION

From 2030 EU law requires passenger ships at berth in core ports to connect [6], which could bring use close to the design potential.

EVIDENCE AGAINST

Until then connection is the shipping line's choice; the law has exemptions (for example grid stability).

WHY THEY DIFFER, AND WHAT IT MEANS HERE

For this claim: a future 90% depends on the 2030 obligation, not on completing the project.

6

Testing the claim

Sub-claim	What the evidence shows	Rating
A. Shore power cuts a connected ship's emissions at berth by about 90%	True for NOx, PM and SOx per connected liner (NO2 –93%, PM –92.6%, SO2 –99.6%) [10, 11]; CO2 falls only about 40%; consistent with research [7, 9]. Infrastructure Malta's own study is not public.	LARGELY SUPPORTED
B. The project slashes 90% of air pollution in the Grand Harbour	At most about 9% of at-berth cruise emissions in year one; other sources and manoeuvring untouched; no harbour measurement.	NOT SHOWN
C. 17,000 families benefit from cleaner air	Basis of the figure not found; any benefit scales with use.	NOT SHOWN
D. EUR 33 million; 17 points; five liners at once	17 points and five liners consistent [1–3]. Cost figures vary by phase and date: EUR 49.9m for the whole project and EUR 37m for phase 1 (2020) [10, 11]; EUR 33m at the 2024 launch [2].	FIGURES VARY

7

Verdict and requests for evidence

VERDICT

MISLEADING

A design potential presented as a result; connection is voluntary until 2030 and first-year use was 9% of berth time.
Confidence: moderate.

Why. (1) The 90% is Infrastructure Malta’s figure for ships that plug in; its earlier pages limited it to the pollution ships produce [10, 11]. (2) The 2023 project page presented it as a cut in Grand Harbour air pollution and the launch as a result of completing the project, neither saying that connection is voluntary until 2030 [1, 2, 6]. (3) Transport Malta’s own records, as reported, show ships plugged in for 9% of berth time in the first year [4], so the cut was roughly a tenth of the figure claimed. The connection points and capacity are as stated; the overall impression is not. Our scale calls this *Misleading*.

Confidence is moderate because the connection records are known through a news analysis, not the FOI reply itself, and no air-quality measurement exists either way.

Evidence we are asking for

- ☐ Transport Malta’s OPS connection log by call and hours, July 2024 to date.
- ☐ The study behind the per-pollutant cuts and the 17,000-family figure.
- ☐ The results of ERA’s Senglea shipping study (mobile station, until October 2024).
- ☐ The shore power agreements’ terms on use, which Transport Malta declined to release as commercial.

8

Limitations

- Connection figures are second-hand (Amphora Media’s analysis of a Transport Malta FOI reply) ♦.
- The 9% upper bound assumes zero local emissions while plugged in and the same mix of ships plugged and unplugged; larger ships plugging in could make the cut in tonnes somewhat larger.
- Uptake may have risen since July 2025; no later figures were found.
- The per-pollutant cuts are Infrastructure Malta’s own estimates; the study behind them was not found.

- Cruise calls are calendar years; the uptake figures cover July 2024 to July 2025.
- Emissions shifted to the power station and the interconnector were not estimated.
- Whether MSC World Europa burned LNG or marine gasoil while berthed in Valletta is not known; its fuel is described by its owner [12].
- The counts reconcile ($312 + 38 + 23 = 373$ berths; $61 + 6 = 67$ connections), but Amphora Media's 16.2% for stays of one to two days does not divide 38 exactly (6 of 38 is 15.8%).
- Peer-reviewed papers were read as abstracts.

References

- [1] Infrastructure Malta (1 Dec 2023). Shore-to-Ship. Project page. <https://www.infrastructuremalta.com/projects/shore-ship>
- [2] TVMnews (10 Jul 2024). Grand Harbour shore-to-ship project completed, to cut down emissions by 90%. <https://tvmnews.mt/en/news/shore-to-ship-project-in-grand-harbour-completed-to-cut-down-emissions-by-90/>
- [3] MaltaToday (10 Jul 2024). Grand Harbour's shore-to-ship project launched. https://www.maltatoday.com.mt/news/national/130180/grand_harbours_shoretoship_project_launched
- [4] Demarco J. (17 Oct 2025). Cruise ships plugged in Grand Harbour onshore power supply only 9% of time in first year. *Amphora Media*. ♦ Transport Malta FOI data. <https://www.amphora.media/2025/10/cruise-ships-shore-to-ship-power-malta>
- [5] Valletta Cruise Port (2026). Valletta Cruise Port closes 2025 with strong growth and nearly one million passenger movements. <https://www.vallettacruiseport.com/news-media/valletta-cruise-port-closes-2025-with-strong-growth-and-nearly-one-million-passenger-movements>
- [6] Regulation (EU) 2023/1805 (FuelEU Maritime), Article 6. OJ L 234, 22.9.2023. <https://eur-lex.europa.eu/eli/reg/2023/1805/oj>
- [7] Mocerino L., Ergin S., Temren G. (2026). Emission inventory of cruise ship exhaust emissions at Istanbul Galataport (2024). *Atmosphere* 17(4):360. doi:10.3390/atmos17040360. (Abstract read.) <https://doi.org/10.3390/atmos17040360>
- [8] Gagić R., Škuric M., Djukanović G., Nikolić D. (2022). Establishing correlation between cruise ship activities and ambient PM concentrations in the Kotor Bay area. *Atmosphere* 13(11):1819. doi:10.3390/atmos13111819. (Abstract read.) <https://doi.org/10.3390/atmos13111819>
- [9] Rogošić M., Stanivuk T., Lucaci D. (2025). A study on the application of shore-side power as a method to reduce the emissions of greenhouse gases by cruise ships. *J. Marine Science and Engineering* 13(3):453. doi:10.3390/jmse13030453. (Abstract read.) <https://doi.org/10.3390/jmse13030453>
- [10] Infrastructure Malta (30 Nov 2020). Grand Harbour Clean Air Project gets underway. Read 5 Oct 2026. <https://www.infrastructuremalta.com/news/grand-harbour-clean-air-project-gets-underway>
- [11] Infrastructure Malta (19 Feb 2022). First Grand Harbour Clean Air Project frequency converters in place. Read 5 Oct 2026. <https://www.infrastructuremalta.com/news/first-grand-harbour-clean-air-project-frequency-converters-place>
- [12] MSC Cruises (24 Oct 2022). Sustainable cruising: the Cruise Division of MSC Group and Chantiers de l'Atlantique mark a major step forward with the delivery of MSC World Europa. Press release; also 12 Apr 2023 (Valletta in its itineraries). Read 5 Oct 2026. https://www.msypressarea.com/en_US/press-releases/sustainable-cruising-the-cruise-division-of-msc-group-and-chantiers-de-l-atlantique/
- [13] Mižien. Data and calculations: data/cc-016/; tools/cc-016-report/calc.py.

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	3 Oct 2026	First issue. Right of reply to Infrastructure Malta and Transport Malta not yet sent.
1.1	5 Oct 2026	Corrections: (1) “No source gives the basis of the 90%” / “basis not published” → Infrastructure Malta’s 2020 and 2022 pages give it, scoped to the pollution ships produce, with per-liner cuts (NO ₂ –93%, PM –92.6%, SO ₂ –99.6%, CO ₂ –39.6%) [10, 11]; the 2023 page dropped the scope (Section 1). (2) Section 3: the 90% inferred from hotelling’s share of CO ₂ [7] → Infrastructure Malta’s own per-pollutant figures, research kept as context. (3) Sub-claim D “Consistent” → “Figures vary”: EUR 33m (2024 launch), 37m (phase 1) and 49.9m (whole project); the TL;DR no longer attributes EUR 33m to the project page, which gives no cost. (4) TL;DR point 4 “extra calls roughly cancel the emissions saved” → states that calls are calendar years, uptake July–July and shore power was available from July 2024. (5) 2019 calls: also given by Infrastructure Malta citing the NSO. ‘What would move it down’: removed the clause saying no scoped wording was found. Verdict and confidence unchanged.
1.2	5 Oct 2026	Upgrade: (1) New Figure 1: the 373 berths split into 35 plugged in by MSC World Europa, 32 by all other ships and 306 not plugged in, beside the 90% per connected liner and the 9% of berth time plugged in (counts re-read in [4] and reconciled; second-hand); former Figures 1 and 2 renumbered 2 and 3. (2) MSC World Europa is LNG-powered (dual-fuel LNG / low-sulphur marine gasoil), from its owner’s release [12]: Section 3, Q1 and limitations. (3) Plug-in rates added: MSC World Europa 35 of 51 calls, all other ships 32 of 322. (4) TL;DR point 3, a tile (306 berths not plugged in), Q2 and flyer updated. Verdict and confidence unchanged.
1.2	5 Oct 2026	Maintainer decision (5 Oct 2026): sub-claim A rated Largely supported (was Plausible); its evidence text and the Q1 label updated to match.